MEDMIND

"Your Personalized Support for Meds, Mood, and Mind."

LUDDY
SCHOOL OF INFORMATICS,
COMPUTING, AND ENGINEERING
Indianapolis

AGENDA

WHAT IS

MEDMIND ?

MedMind is a smart mobile health platform designed to support individuals with mood disorders such as depression, anxiety, and bipolar disorder. Focused on improving medication adherence and emotional engagement, MedMind combines smart medication reminders, pharmacy refill tracking, and AI-driven mood journaling by integrating passive behavioral data from wearable devices, offering a comprehensive view of a user's mental health, and helping both patients and providers manage care more effectively.

WHO ARE WE

A team with a passion for ethical, inclusive mental health innovation.

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THE PROBLEM

Medication Non- Adherence Crisis

40-60% of psychiatric patients fail to take medications as prescribed, leading to relapse, and increased risk.

Stigma & Privacy Concerns

Patients fear judgment and disclosure, avoiding treatment or discontinuing medications silently rather than discussing struggles with providers.

Memory & Motivation Barriers

Mental health conditions impair memory and motivation, making traditional reminders ineffective.

Fragmented Care Systems

Existing tools lack integration with clinical workflows, offering no real-time feedback and support.

Lack of Provider Visibility

Infrequent appointments and self-reporting limit psychiatrists' ability to identify issues and intervene proactively.

Gap Between Appointments

Care is episodic (visits every 2-4 weeks), leaving patients unsupported during critical moments.

PROPOSAL

THE PROPOSAL TO SOLUTION

OBJECTIVES

- Develop a mobile application that increases psychiatric medication adherence by at least 30%
- Implement a real-time provider dashboard that allows providers to track metrics.
- Add a secure AI chat for questions and mood journaling

SCOPE

- End-to-end design, development, and pilot testing of mobile application.
- Improve medication adherence and emotional engagement
- Provide insight to providers
- Core components
- Smart medication reminders
- Adherence logging
- Pharmacy refill tracking
- Wearable compatibility
- AI chatbot
- Provider features
- Dashboard for adherence data, mood trends
- Notifications of risky trends

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PROJECT PROPOSAL

PROJECT PHASES

Timeframe

	Description of Work	Start and End Dates
Phase One	Planning and Requirement Gathering	Sept 15 - Sept 30
Phase Two	Design and Development	Oct 1 - Nov 15
Phase Three	Pilot Testing,Evaluation and Product Launch	Nov 16 - Nov 30

Project Budget

	Description of Work	Anticipated Costs
Phase One	1. User research and focus group with patients and providers	$15,000
	2. Clinical and ethical consultations	$10,000
	3. UX/UI Design and prototyping	$15,000
	4. Technical architecture & security planning	$8,000
	5. Project management & stakeholder workshops	$7,000
Phase Two	1. Cross-platform mobile app development	$90,000
	2. Backend & API development	$50,000
	3. AI chatbot development & NLP training	$30,000
	4. Wearable device integrations	$15,000
	5. Provider dashboard development	$25,000
	6. Security & compliance implementation	$15,000
	7. Quality assurance & automated testing	$15,000
	8. Cloud infrastructure & maintenance	$12,000
	9. UX refinement & accessibility audits	$10,000

Project Proposal: MedMind

Phase Three	1. Pilot recruitment & participant incentives	$20,000
	2. Data collection, analytics & impact evaluation	$25,000
	3. Marketing & communication campaign	$35,000
	4. App store submission & regulatory certification	$8,000
	5. Customer support setup & training	$20,000
	6. Post-launch monitoring & rapid iteration	$18,000
	7. Documentation & training materials development	$10,000
	Total	**$ 453,000**

PROJECT PROPOSAL

MONITORING AND EVALUATION

Project Monitoring

- Continuous tracking across all phases
- Weekly sprint reviews for development progress
- Biweekly stakeholder meetings for milestones, risks, and adjustments
- Real-time technical monitoring (uptime, response time, integrations)
- Pilot analytics: reminder use, chatbot interactions, adherence logs

Project Evaluation

- Outcome & process evaluation across all objectives
- Objective 1: Medication adherence → target 30% increase
- Objective 2: Provider visibility → 90% report improved insights
- Objective 3: Chatbot engagement → 75% mood journaling ≥3x/week
- Cross-cutting indicators: satisfaction, retention, HIPAA/GDPR compliance

SCOPE

PROJECT DELIVERABLES

WHAT'S INCLUDED

Medmind Mobile Application (iOS & Android)

- A fully functional, cross-platform app for patients including:
- Smart medication reminders
- Daily adherence tracking
- Pharmacy refill alerts
- Mood journaling and emotional check-ins via an AI chatbot
- Wearable integration (e.g., Fitbit or Apple Watch) for passive data collection

Provider Dashboard

- A secure digital platform for clinicians providing:
- Real-time visualization of patient medication logs
- Alerts for missed doses, emotional distress, or risky patterns
- Data analytics for treatment decision

PROJECT DELIVERABLES

WHAT'S INCLUDED

Provider Dashboard

- A secure digital platform for clinicians providing:
- Real-time visualization of patient medication logs
- Alerts for missed doses, emotional distress, or risky patterns
- Data analytics for treatment decision

Pilot Testing & Evaluation Framework

- A structured pilot program including:
- 20–50 participants + 3–5 providers
- Consent forms, onboarding materials, and training guides
- Backend analytics to track engagement, adherence, and mood data
- Surveys, interview protocols, and evaluation tools

PROJECT DELIVERABLES

WHAT'S INCLUDED

Product Launch Package

- All assets and infrastructure needed for public release:
- App Store and Google Play Store submissions
- Promotional materials (website, app store graphics, social media assets)
- Monitoring mechanisms for post-launch usage, engagement, and bugs

Technical & Compliance Records

- Comprehensive documentation including:
- HIPAA/GDPR compliance checklists and audit results
- Privacy policy and data-handling workflows
- User manuals (patients and clinicians)
- Maintenance and technical documentation for future teams

PROJECT DELIVERABLES

WHAT'S EXCLUDED

Mental Health Service Limitations

- Support for all psychiatric conditions (e.g., schizophrenia, personality disorders)
- Live therapy, real-time clinician chat, or emergency crisis intervention services

Technical & Integration Exclusions

- Full EHR/EMR integration with systems like Epic or Cerner
- Insurance system integration, claim processing, or billing
- Multi-language localization or advanced accessibility add-ons
- Predictive AI features, such as risk detection or clinical diagnosis

Expansion Limitations

- Large-scale pilot programs or multi-site implementation
- Additional wearable integrations beyond one initial device
- Advanced personalization or dynamic learning features
- Any new post-launch features beyond the MVP

PROJECT OBJECTIVES

COST OBJECTIVE

- **Budget Cap**: $453,000
- **Phase Variance Allowed**: ±5%
- Weekly cost reviews & corrective actions

Success: Deliver features + pilot testing within $453k

SCHEDULE OBJECTIVE

- **Timeline**: Sept 15 → Nov 30, 2025
- **Milestones :**
- Planning & Requirements : Sept 30
- Design & Development : Nov 15
- Pilot Testing & Launch : Nov 30

Success Metric: 95% of milestones on time

QUALITY OBJECTIVE

Clinical Impact	Provider Experience	Engagement	Reliability
+30% adherence	90% provider visibility	75% chatbot use	99% uptime

ACCEPTANCE CRITERIA

Feature Completion

- All core features built & functioning

Security

- Compliant with HIPAA & GDPR
- Security audit passed (no critical vulnerabilities)

Functionality

- Stable performance across iOS, Android & Web
- No major bugs or crashes

User Acceptance Testing (UAT)

- ≥ 90% user satisfaction
- All high-priority bugs resolved pre-launch

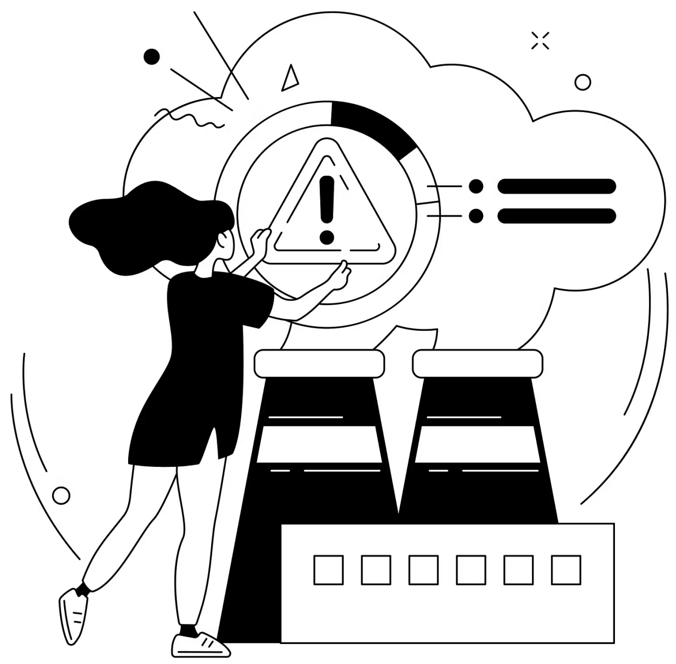
Performance

- < 2s load time for 90% of users
- ≥ 95% uptime (30 days)

Documentation

- User manuals & onboarding guides complete
- Internal documentation for maintenance

CONSTRAINTS



- Fixed 10- week timeline
- Minimal Viable Product (MVP) Delivery Model
- High-cost components must remain within strict limits
- Limited pilot size: 50 patients, 5 providers
- Strict HIPAA/GDPR requirements
- Limited AI chatbot scope (non-clinical)

WBS

(WORK BREAKDOWN STRUCTURE)

WORK BREAKDOWN STRUCTURE

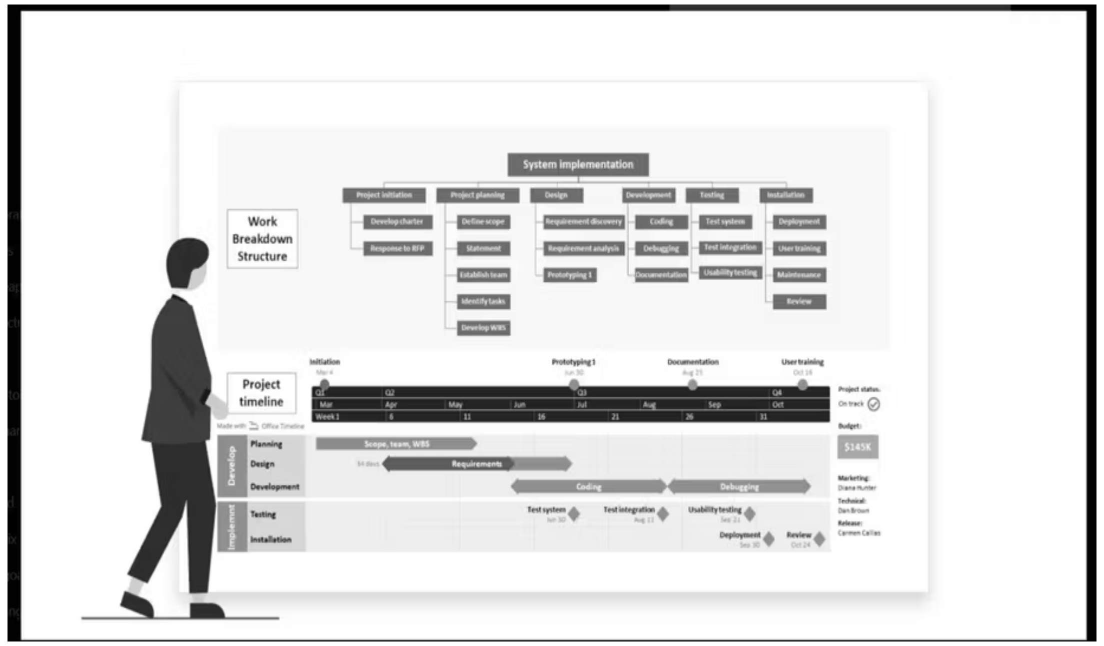
Why We Created This WBS

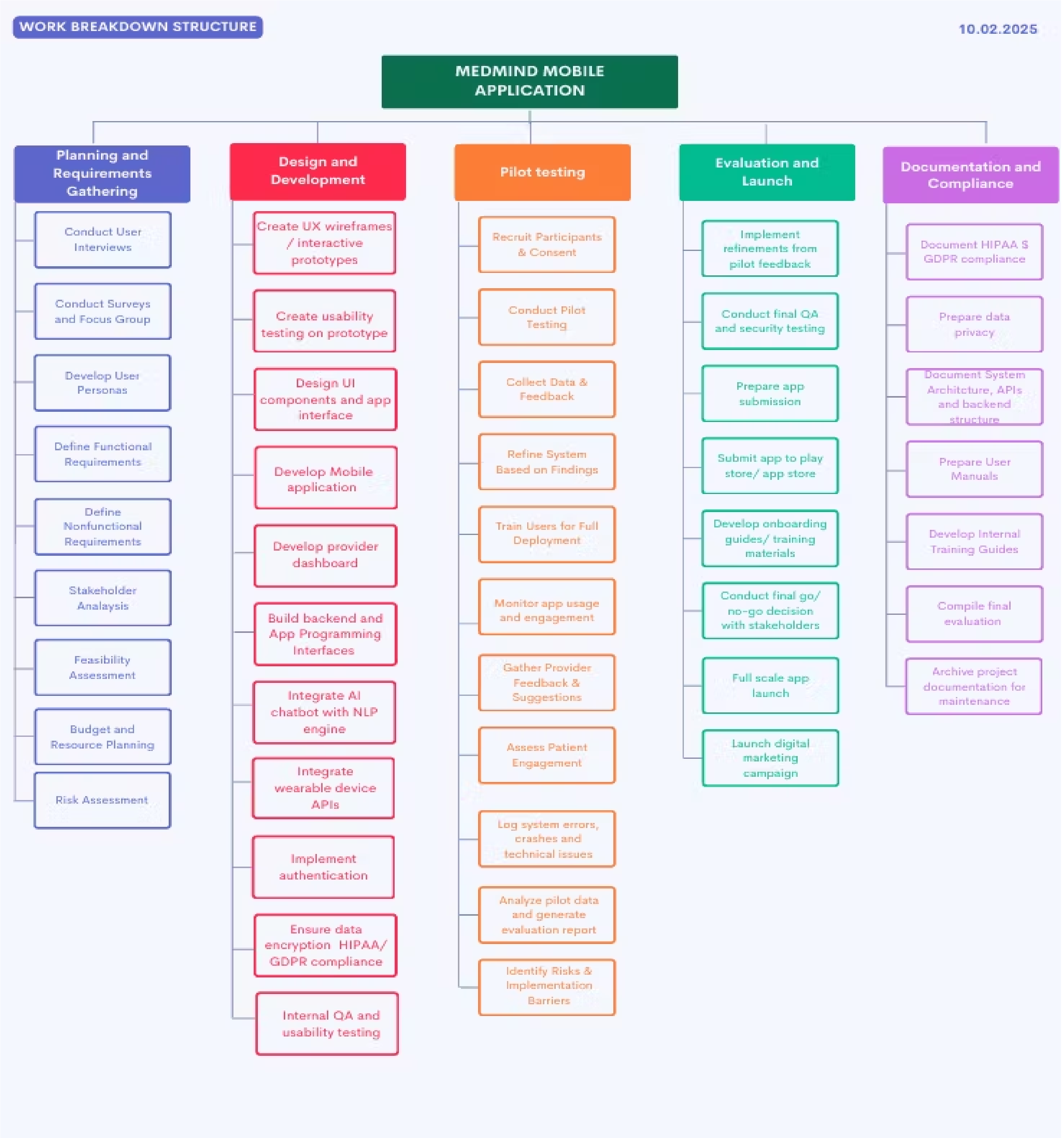
- To organize all tasks needed to build the MedMind mobile application.
- To divide responsibilities among our team members.
- To visualize the flow of work from planning → development → testing → launch → documentation.
- Ensures nothing is missed, especially with HIPAA/GDPR compliance requirements

Overview of Our WBS Structure

Our WBS is divided into five major phases:

1. PHASE 1 - Planning & Requirements Gathering
2. PHASE 2 - Design & Development
3. PHASE 3 - Pilot Testing
4. PHASE 4 - Evaluation & Launch
5. PHASE 5 - Documentation & Compliance




WORK BREAKDOWN STRUCTURE
10.02.2025
MEDMIND MOBILE APPLICATION
Planning and Requirements Gathering
Conduct User Interviews
Conduct Surveys and Focus Group
Develop User Personas
Define Functional Requirements
Define Nonfunctional Requirements
Stakeholder Analaysis
Feasibility Assessment
Budget and Resource Planning
Risk Assessment
Design and Development
Create UX wireframes / interactive prototypes
Create usability testing on prototype
Design UI components and app interface
Develop Mobile application
Develop provider dashboard
Build backend and App Programming Interfaces
Integrate AI chatbot with NLP engine
Integrate wearable device APIs
Implement authentication
Ensure data encryption HIPAA/ GDPR compliance
Internal QA and usability testing
Pilot testing
Recruit Participants & Consent
Conduct Pilot Testing
Collect Data & Feedback
Refine System Based on Findings
Train Users for Full Deployment
Monitor app usage and engagement
Gather Provider Feedback & Suggestions
Assess Patient Engagement
Log system errors, crashes and technical issues
Analyze pilot data and generate evaluation report
Identify Risks & Implementation Barriers
Evaluation and Launch
Implement refinements from pilot feedback
Conduct final QA and security testing
Prepare app submission
Submit app to play store/ app store
Develop onboarding guides/ training materials
Conduct final go/ no-go decision with stakeholders
Full scale app launch
Launch digital marketing campaign
Documentation and Compliance
Document HIPAA & GDPR compliance
Prepare data privacy
Document System Architecture, APIs and backend structure
Prepare User Manuals
Develop Internal Training Guides
Compile final evaluation
Archive project documentation for maintenance

PROJECT BUDGET

PROJECT BUDGET

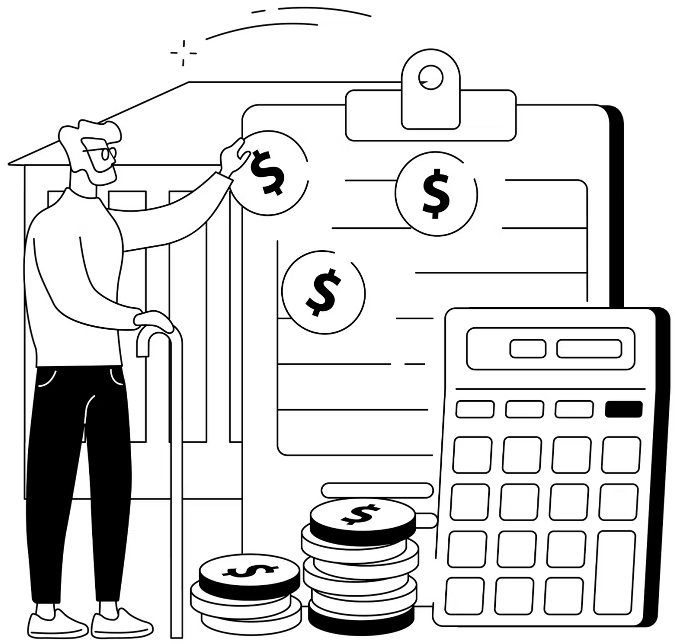
Budget Overview

Total Project Cost: **$453,000**

- Budget distributed across the 5 project phases:
- Planning & Requirements – $55,000
- Design & Development – $245,000
- Pilot Testing – $60,000
- Evaluation & Launch – $63,000
- Documentation & Compliance – $30,000

Emphasize:

- Stayed within the allowed variance
- Supports labor, materials, travel, and compliance activities

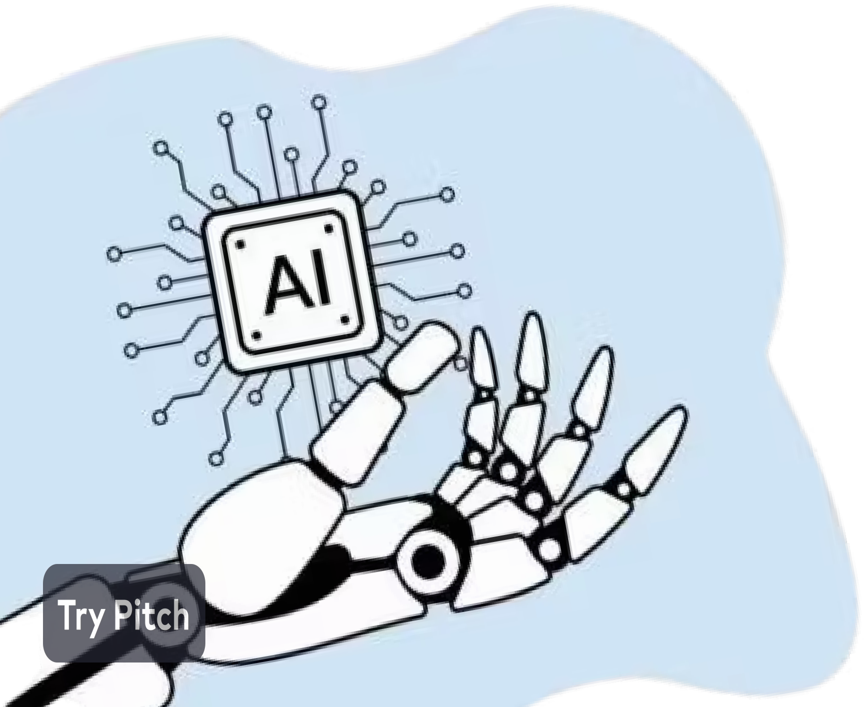


PHASE 1: PLANNING & REQUIREMENTS ($55,000)

In this phase, our goal was to understand the people we're designing for. Most of our budget here went toward conversations — interviewing users, conducting surveys, and running focused discussions to understand their needs. We also spent time shaping personas and defining what the system must do and how it should behave. Because this is the foundation of the entire project, we invested resources into clarity: identifying constraints, analyzing stakeholders, and making sure the project was feasible before moving forward.

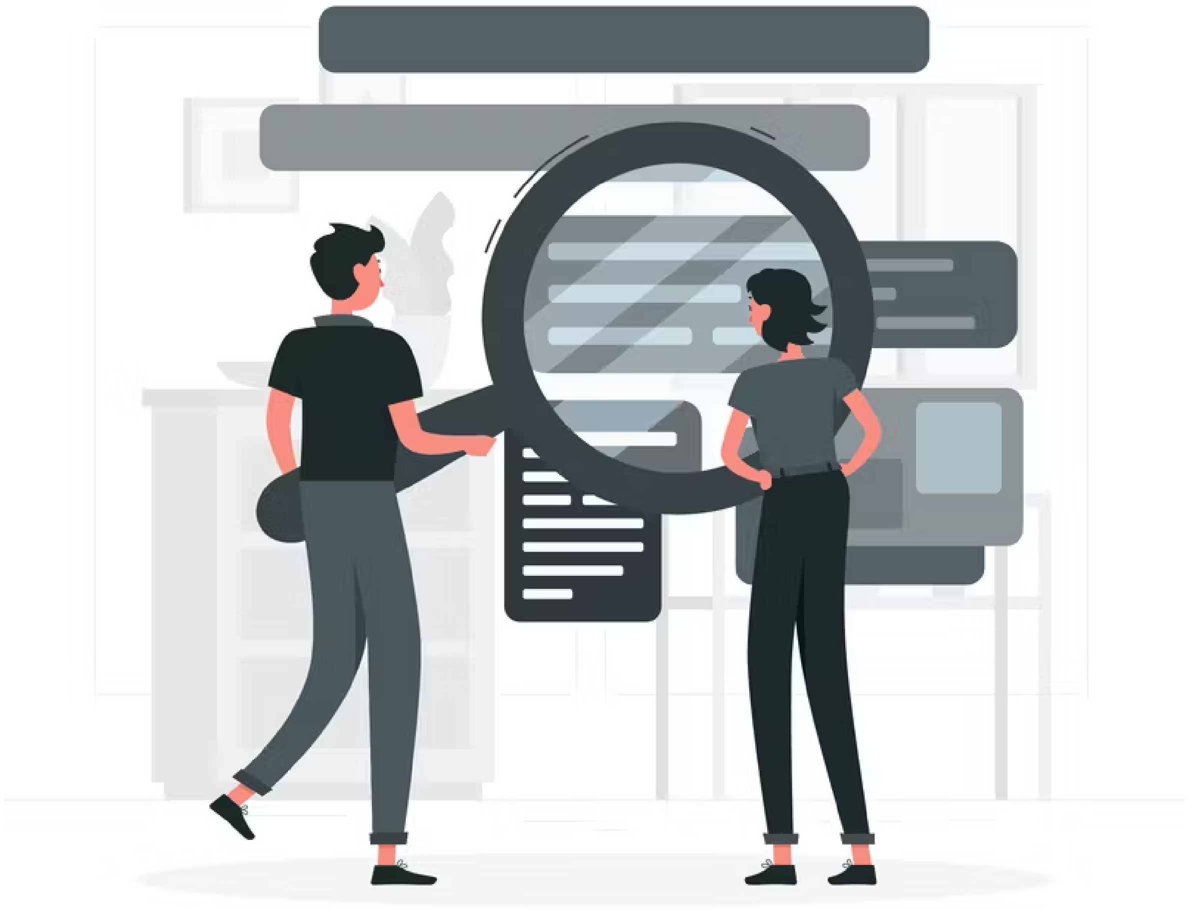
PHASE 2: DESIGN & DEVELOPMENT ($245,000)

This is where most of our project costs appear, and it makes sense — this is the heart of building the actual product. During this phase, we moved from concepts into real, functional experiences. We designed wireframes, refined the UI, tested early prototypes, and then transitioned into full-scale development. A large portion of the budget supported engineering work: building the mobile app, creating the provider dashboard, integrating AI chat capabilities, securing data, and making sure the backend could support everything. This phase captures the transformation from idea to working system.



PHASE 3: PILOT TESTING ($60,000)

After development, we needed to see how the system performs in real life. This phase focused on getting actual users to try the app and share their honest feedback. Our budget here supported participant recruitment, guided pilot sessions, and detailed observation of how people interacted with the system. We monitored what worked, where confusion happened, and what needed improvement. This is where we strengthened the product — not by guessing, but by learning directly from the users themselves.



PHASE 4: EVALUATION & LAUNCH ($63,000)

With insights from the pilot in hand, we refined the system and prepared it for launch. This phase includes final rounds of QA, making sure the app meets security and performance expectations, and getting it ready for the app stores. A portion of the budget supported onboarding materials and launch communication to ensure users knew how to navigate and benefit from the app. Once everything aligned, we initiated the full rollout, supported by a small but targeted marketing effort.

PHASE 5: DOCUMENTATION & COMPLIANCE ($30,000)

The final phase ensures the project is complete not only functionally but also responsibly. This phase involved documenting how the system works internally, creating user guides, and preparing compliance documentation for HIPAA and GDPR. We organized everything from training materials to technical architecture details so the system can be maintained and built upon in the future. This phase ensures the app isn't just launched — it's sustainable and compliant.

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WEEKLY & CUMULATIVE COST CHARTS

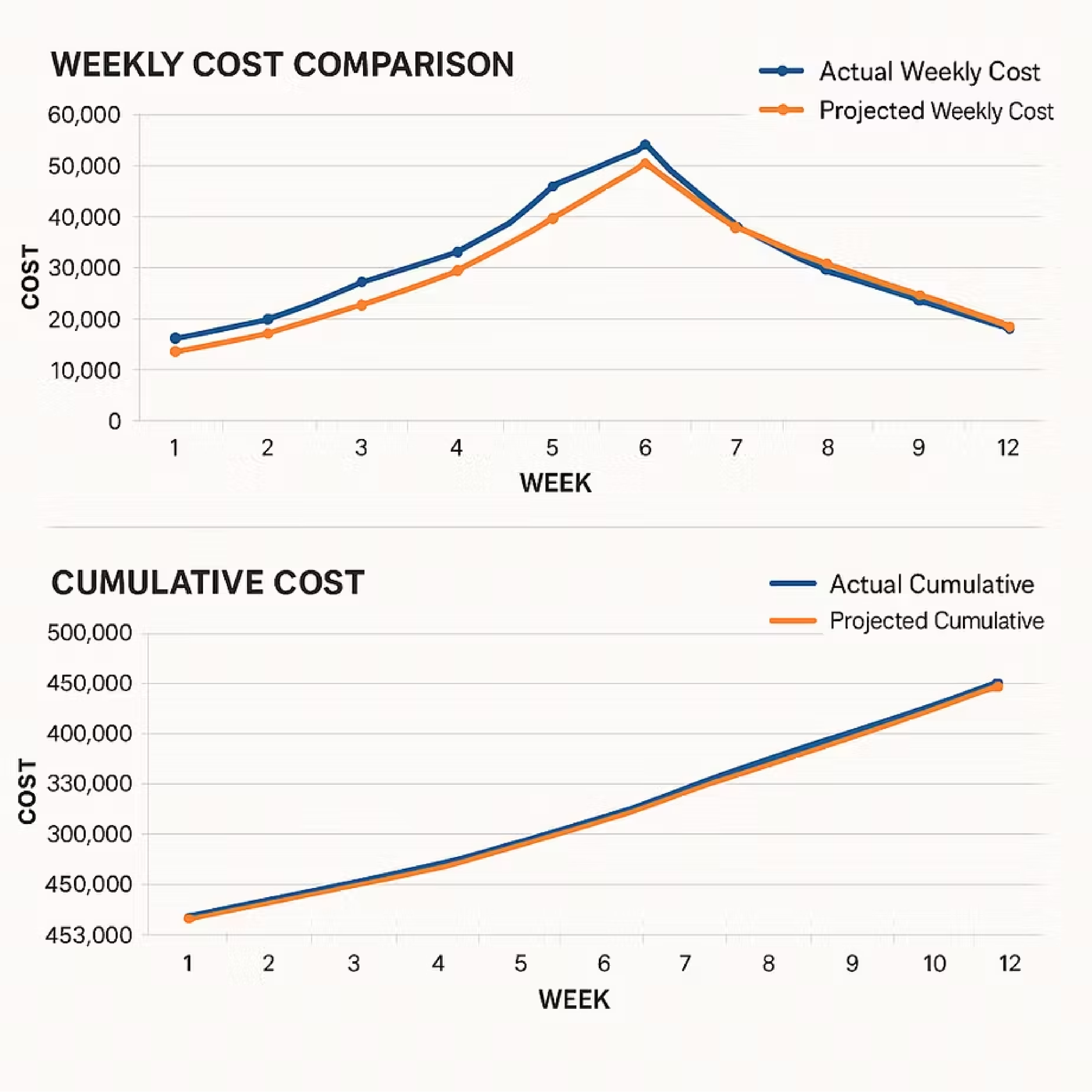
Weekly Cost Comparison

- Actual vs. projected
- Clear peak around development phase

Cumulative Cost Chart

- Actual stays close to projected
- Ends exactly at $453,000

These visuals demonstrate cost control, variance management, and stable project spending.



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BUDGET SUMMARY

- All five phases completed within budget
- Labor hours, material, travel, and other costs fully documented
- Actual cost equals projected cost: $453,000
- Budget supports full delivery of the MedMind application

"Our project maintained financial stability while supporting complex technical development, user testing, and compliance requirements."

PROJECT WORK PLAN

WORK PLAN

A work plan is a detailed roadmap of project execution

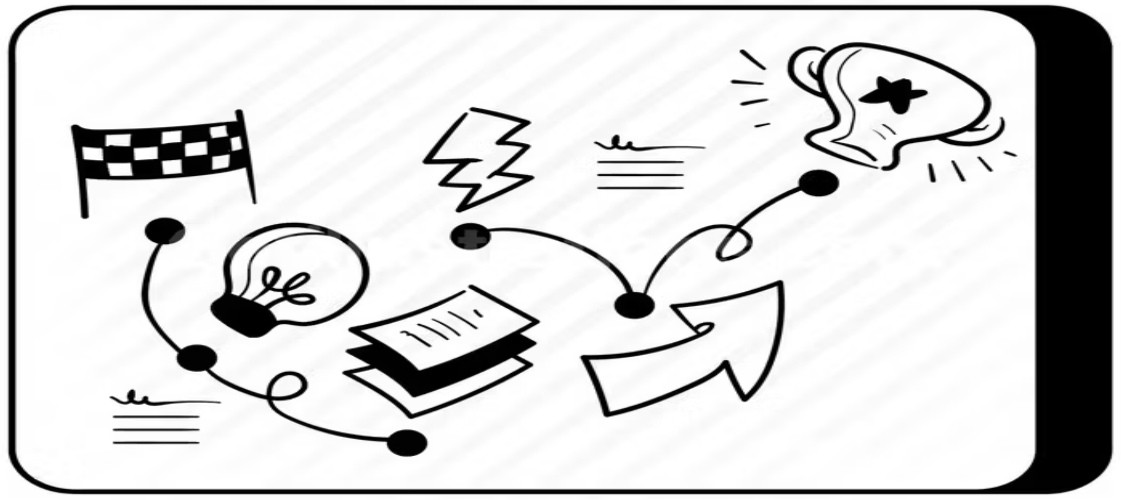
It outlines tasks, timelines, responsibilities, and dependencies

It ensures the project stays on schedule and within scope

The work plan outlines tasks, owners, timelines, and dependencies across 5 project phases:

- Project Planning & Requirement Gathering
- Design & Development
- Pilot Testing
- Evaluation & Launch
- Documentation & Compliance

Timeline: September 15 – November 28, 2025 (11 weeks)



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WORK PHASES

Breaking the project into phases helped our team stay organized, reduce risks, and successfully move from idea to launch.

Five Phases:

1. **Planning**: Decide what to build
2. **Design & Development**: Build the app
3. **Pilot Testing:** Test with real users
4. **Evaluation & Launch:** Fix, approve, and release
5. **Documentation:** Record everything for future use

- Each phase depends on the previous one
- Skipping phases increases risk
- Work phases help track progress
- Work phases improve teamwork
- Work phases reduce confusion

PHASE 1: PROJECT PLANNING

This is the thinking and preparation phase.

“Before building anything, we first understand the problem and what users need.”

If planning is weak, the whole project can fail.

Duration: Sept 15–30, 2025

Objectives: Understand users, define system needs, assess feasibility.

Key Tasks:

- User interviews
- Surveys & focus groups
- User personas
- Functional requirements
- Nonfunctional requirements
- Stakeholder analysis
- Feasibility, resources, budgeting, risks

PHASE 2: DESIGN & DEVELOPMENT

This is the building phase.
“Here we design how the app looks and actually start developing it.”

Duration: Oct 1–Nov 15, 2025

Key Tasks:

- UX wireframes & prototypes
- Usability testing on prototype
- UI components & internal testing
- Mobile app development
- Provider dashboard
- Backend interfaces
- AI chatbot with NLP
- Wearable device integration
- Authentication & encryption (HIPAA/GDPR)
- Internal QA testing

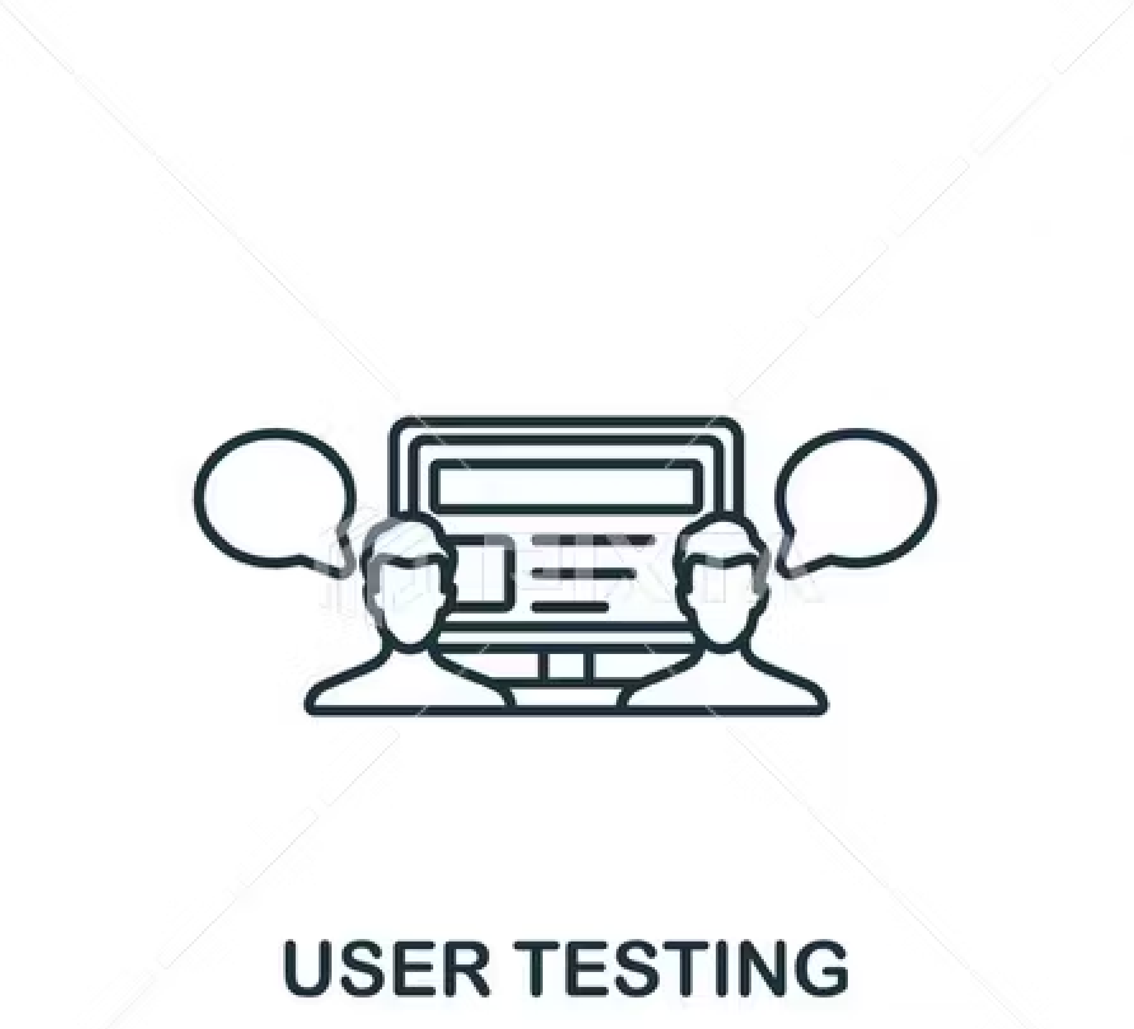
PHASE 3: PILOT TESTING

This is the testing with real users phase.
“We let real users try the app to see what works and what doesn’t.”

Duration: Nov 15–28, 2025

Key Tasks:

- Recruit participants & obtain consent
- Conduct pilot tests
- Collect data & user feedback
- Refine system based on results
- Train end users
- Monitor usage & engagement
- Provider feedback
- Patient engagement assessment
- Error logging & technical issue tracking
- Pilot evaluation report



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PHASE 4: EVALUATION & LAUNCH

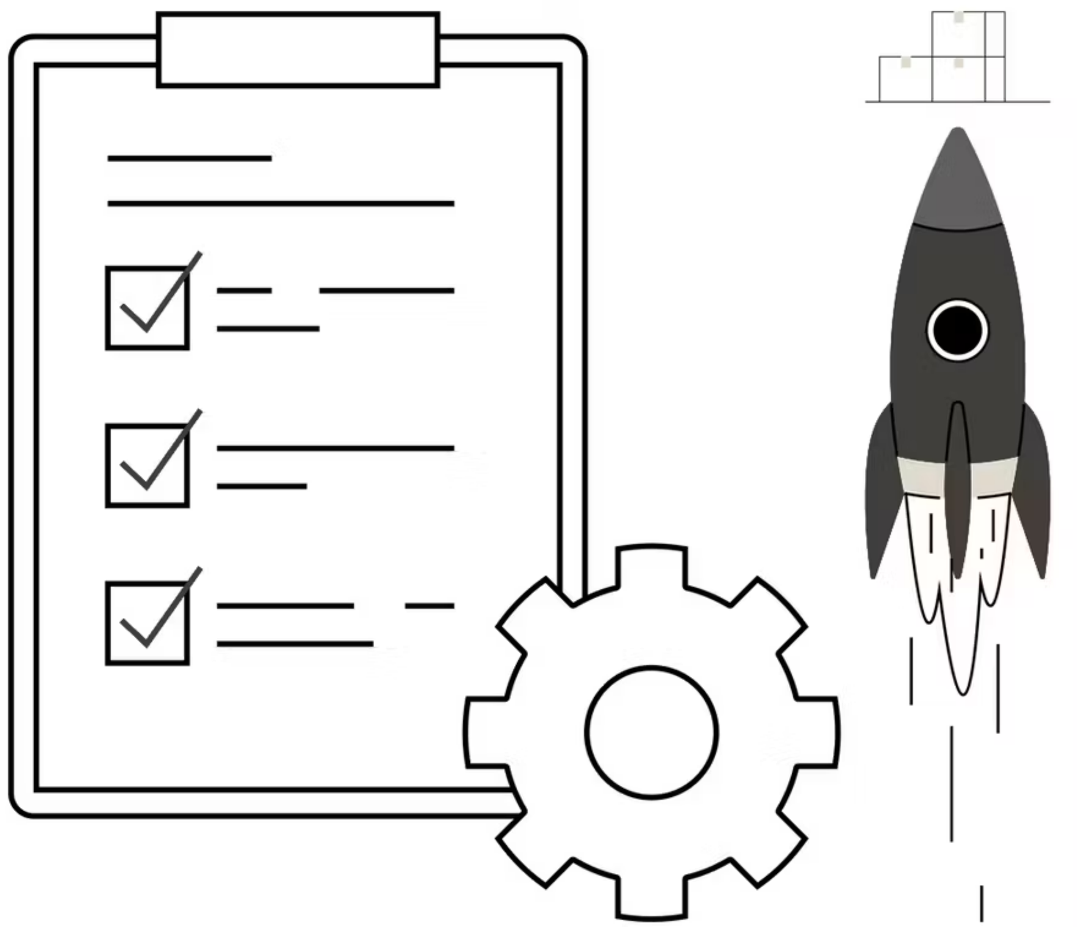
This is the final check and release phase.

"We fix issues, get approvals, and launch the app for everyone."

Duration: Nov 24–28, 2025

Key Tasks:

- Implement refinements from pilot
- Final QA & security test
- Prepare and submit app
- Go/No-Go stakeholder decision
- Full-scale app launch
- Digital marketing campaign



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PHASE 5: DOCUMENTATION & COMPLIANCE

This is the record-keeping and future support phase.

“We document everything so the app can be maintained and trusted.”

Duration: Nov 24–28, 2025

Key Tasks:

- HIPAA/GDPR documentation
- Data privacy documentation
- System architecture & API documentation
- User manuals
- Internal training guides
- Final evaluation compilation
- Archiving project files

COMMUNICATION PLAN

COMMUNICATIONS PLAN

Importance of Communication

- Effective communication is critical for the success of our MedMind project because it ensures all team members and stakeholders are aligned on project goals, progress, and risks.
- Clear communication supports collaboration between development, design, research, and quality assurance teams, helps maintain regulatory compliance, and ensures that decision-makers can act promptly based on accurate information.

COMMUNICATION OBJECTIVES

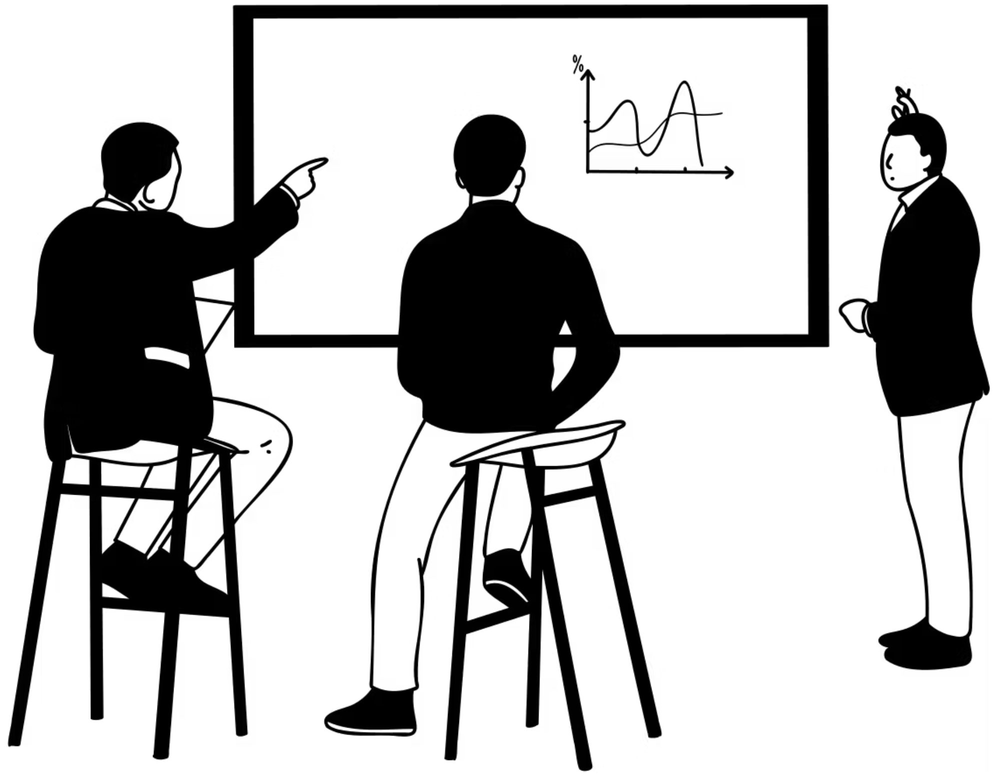
The main objectives of the communication plan are to keep all team members and stakeholders informed, facilitate efficient collaboration across teams, provide timely updates to sponsors and clients, and maintain thorough documentation to support transparency and accountability.

PRE-PROJECT COMMUNICATIONS

- Before the project began, our team leader formally announced the project approval, shared the vision, introduced stakeholders, and confirmed team roles.
- A kickoff meeting aligned expectations, reviewed objectives, success criteria, and communication norms.
- Roles and responsibilities were clarified, the project charter was distributed, and collaboration tools such as Microsoft Teams, Canvas, Jira, Figma, Google Drive, GitHub, and email were set up to ensure smooth coordination.

PHASE 1 – PLANNING AND REQUIREMENTS

- Phase 1 focused on understanding user needs and defining project requirements.
- Workshops were conducted to develop and validate user personas.
- Risk assessment workshops identified potential issues, and a Phase 1 completion report confirmed readiness to proceed to design and development.



PHASE 2 – DESIGN AND DEVELOPMENT

- Phase 2 involved active development and frequent collaboration.
- Sprint planning meetings organized tasks and reviewed core features such as medication reminders, mood tracking, and chatbot functionality.
- Security reviews and QA reporting maintained compliance and quality standards, while code merges and version control kept the system stable.

PHASE 3 – PILOT TESTING.

- Pilot testing validated the application with real users and healthcare providers.
- Participants were recruited through emails, flyers, social media, and referrals.
- Providers received training on the dashboard, and participants were given detailed instructions and support.

LAUNCH AND CLOSURE

- The go/no-go meeting with the executive team finalized launch decisions, app store submission was tracked, and data privacy impact assessments ensured compliance.
- Following the successful launch, the comprehensive project closure report documented achievements, lessons learned, and recommendations for future projects.
- Finally, a recognition event celebrated team contributions and reinforced a positive project culture.

COMMUNICATION TOOLS

- **Microsoft Teams** for meetings and day-to-day coordination
- **Canvas** for academic collaboration and content sharing
- **Jira** for sprint planning and task tracking
- **Figma** for design communication and prototyping
- **Google Drive** for shared documentation
- **GitHub** for version control and code collaboration
- **Email** for formal stakeholder updates

BENEFITS OF THE COMMUNICATION PLAN

Our structured communication plan prevents misunderstandings, improves accountability, supports compliance, and enables informed decision-making. It strengthens collaboration across teams, reduces risks, and ensures the successful delivery of our MedMind project.

RISK ANALYSIS

WHAT IS RISK ANALYSIS?

1. **Identify** potential problems that could happen during the project
2. **Measure** how serious each problem could be (Likelihood × Impact)
3. **Plan** solutions to prevent or reduce problems
4. **Monitor** risks throughout the project lifecycle

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PROBLEMS ENCOUNTERED DURING DEVELOPMENT

1. API Integration Failure (RPN: 80)

Mobile app, backend, wearables, and chatbot systems couldn’t communicate

Challenge: Inconsistent data formats, A mismatches

2. Security Vulnerability (RPN: 75)

Patient health data (PHI) at risk without proper encryption

Challenge: HIPAA/GDPR compliance, secure token handling

3. Chatbot Logic Errors (RPN: 64)

AI chatbot generating medically inappropriate responses

Challenge: Incomplete NLP training, missed critical symptoms

HOW WE SOLVED THESE CHALLENGES

1. Technical Solutions: Clear API documentation, incremental testing, automated test suites, security audits with penetration testing, OWASP scanners

2. Team Process: Daily standups, peer code reviews, weekly risk reviews, clear ownership assignments, MoSCoW prioritization

3. Continuous Monitoring: Regression testing, 60% unit test coverage, performance profiling, environment stability checks, automated CI/CD

CHALLENGES & HOW WE OVERCAME THEM

1. **Challenge** : Sprint deadline pressure with complex API integration
   Solution: Added buffer time, daily standups to flag blockers early, strict scope control

2. **Challenge** : Wearable device data delays and inconsistencies
   Solution: Data normalization layer, caching mechanism, simulation mode for testing

3. **Challenge** : Critical bugs during sprints causing delays
   Solution: Weekly regression testing, 60% test coverage, peer code review, hotfix branch

4. **Challenge** : UI/UX usability with elderly patients
   Solution: Usability testing with target users, WCAG 2.1 accessibility guidelines, guided onboarding

PROJECT SUCCESSFULLY COMPLETED!

KEY ACHIEVEMENTS:

1. 2 Critical Risks Closed: Security & UX fully resolved
2. 7 Major Risks Mitigated: Active solutions in place
3. All Systems Integrated: API, backend, wearables, chatbot working together
4. HIPAA Compliant: Patient data secure with encryption & MFA
5. Ready for Pilot Phase: Real-world testing with actual users

CLOSING STATEMENT: "RISK MANAGEMENT IS THE FOUNDATION OF SUCCESSFUL HEALTHCARE SOFTWARE DEVELOPMENT."

PROJECT CONCLUSION

1. Mission Accomplished: MEDMIND healthcare app successfully developed with zero security breaches, full HIPAA compliance, and ready for real-world patient testing

2. Risk Management Success: 20 risks identified, 2 closed, 7 actively mitigated, 7 monitored. Zero critical incidents. 100% on-time delivery

3. Team Excellence: Cross-functional collaboration with daily standups, peer reviews, and shared ownership. Clear communication prevented escalations

4. Key Learnings: Early risk identification, proactive mitigation, and continuous monitoring are the foundation of successful healthcare software development

5. Next Phase: Pilot testing with real patients, gather feedback, iterate, prepare for production launch with full confidence in quality and safety

__MEDMIND__ won’t fix everything.

But it will change the lives it touches and that’s where transformation begins.

THANK YOU !

LUDDY
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